

A Rare Case of Tail Bifurcation in *Ophisops elegans* from Western Anatolia

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Abstract: Tail autotomy is a common and highly effective defensive mechanism observed in many lizard species, allowing individuals to escape predators by shedding a portion of their tail. This behavior, while beneficial for survival, can sometimes result in typical outcomes during tail regeneration. One such rare disorder is tail bifurcation often due to incomplete or misdirected regenerative processes. On June 23, 2025, during fieldwork in the rural areas of Afyonkarahisar Province, a female snake-eyed lizard (*Ophisops elegans*), belonging to the Lacertidae family, was observed with a clearly bifurcated tail. The specimen was captured and examined briefly, and morphometric, biotic, and environmental data were recorded. Photographs were taken to document the anomaly, and the specimen was subsequently released in its habitat. This field observation contributes to the growing number of bifurcated tail reports in Lacertidae and emphasizes the value of in situ herpetological monitoring. Observations in this study enhance our understanding of caudal autotomy and tail regeneration and also provide insights into developmental biology, wound healing, and ecological implications of tail malformations in natural populations.

Keywords: Caudal autotomy, tail regeneration, Lacertidae, field observation, snake-eyed lizard.

Batı Anadolu'dan *Ophisops elegans*'ta Nadir Bir Kuyruk Çatallanması Vakası

Öz: Kuyruk ototomisi, birçok kertenkele türünde gözlemlenen yaygın ve son derece etkili bir savunma mekanizmasıdır. Bu mekanizma, bireylerin kuyruklarının bir kısmını bırakmaları sayesinde yırtıcılardan kaçmalarına olanak tanır. Hayatta kalma açısından faydalı olsa da bu davranış, bazen kuyruk yenilenmesi sürecinde alışılmadık sonuçlara yol açabilir. Bu nadir düzensizliklerden biri, eksik ya da yanlış yönlendirilmiş yenilenme süreçleri sonucunda ortaya çıkan kuyruk çatallanmasıdır. 23 Haziran 2025'te, Afyonkarahisar ilinin kırsal bölgelerinde gerçekleştirilen arazi çalışmaları sırasında, Lacertidae familyasına ait dişi bir tarla kertenkelesi (*Ophisops elegans*) bireyinde belirgin şekilde çatallı bir kuyruk gözlemlenmiştir. Örnek kısa süreliğine yakalanarak incelenmiş; morfometrik, biyotik ve çevresel veriler kaydedilmiştir. Anomaliyi belgelemek amacıyla çeşitli fotoğraflar çekilmiş ve birey, yakalandığı habitatta tekrar doğaya salıverilmiştir. Bu arazi gözlemi, Lacertidae familyasında çatallı kuyrukla ilgili sınırlı fakat artan sayıda raporlara katkı sağlamakta ve yerinde (in situ) herpetolojik gözlemlerin önemini vurgulamaktadır. Bu tür gözlemler, yalnızca kuyruk ototomisi ve kuyruk yenilenmesi hakkındaki bilgilerimizi geliştirmekle kalmaz, aynı zamanda doğal popülasyonlarda kuyruk malformasyonlarının gelişim biyolojisi, yara iyileşmesi ve ekolojik etkileri konusunda da önemli içgörüler sunar.

Anahtar kelimeler: Kaudal ototomi, kuyruk rejenerasyonu, Lacertidae, saha gözlemi, tarla kertenkelesi.

Tail autotomy is a widespread and highly specialized antipredator adaptation among lizards, characterized by the voluntary detachment of the caudal appendage in response to external threats. This self-amputation occurs along pre-formed autotomy planes, either within vertebrae (intravertebral) or at the intervertebral joints, and allows rapid escape when the animal is seized by a predator (Sorlin et al., 2019; Pafilis et al., 2005). The autotomized tail, exhibiting spontaneous movement post-detachment, serves as a distraction mechanism against the predator, thereby increasing the likelihood of survival. Despite its adaptive value, caudal autotomy causes substantial costs such as reduced reproductive success, decreased mobility, and the metabolic demands associated with tissue regeneration (Baban et al., 2022; Bateman & Fleming, 2009; Clause et al., 2006).

The regeneration process is sported by the tail's distinctive morphological and microstructural organization. Notably, the presence of regularly aligned micro-columns with nanoporous caps—structurally

analogous to dense, mushroom-shaped projections—facilitate effective cleavage and subsequent regrowth at the autotomy site within the tails (Baban et al., 2022). Additionally, rapid occlusion of vasculature in the region prevents significant hemorrhage following autotomy (Bülbül & Sarıkurt, 2022).

However, deviations from the typical regenerative process can occur, particularly in cases of incomplete tail regeneration. Under such circumstances, aberrant growth may lead to bifurcation, whereby a secondary tail structure develops along a misaligned axis relative to the original. This condition, which is classified as a morphological anomaly, may alter biomechanical function. This can compromise the lizard's adaptability and survival by negatively affecting vital activities such as its balance during movement, foraging, mating, and escaping from predators. (Passos et al., 2014; Sorlin et al., 2019; Bülbül & Sarıkurt, 2022).

In this study, a case of bifurcation observed and

reported in a specimen of *Ophisops elegans* (Snake-eyed lizard; Ménétries, 1832) during routine fieldwork in Afyonkarahisar province.

Ophisops elegans is a member of the family *Lacertidae* and it shows distribution across a wide range including Türkiye, Armenia, Bulgaria, Greece, the Aegean Islands, Cyprus, Azerbaijan, Iran, Iraq, Syria, Lebanon, Egypt, Pakistan, Jordan, and Israel. Within the Anatolian region, six different subspecies are known (Bozkurt et al., 2022).

On June 23, 2025, during fieldwork conducted under the permit no. E-21264211-288.04-13202316 issued by the Republic of Türkiye Ministry of Agriculture and Forestry, General Directorate of Nature Conservation and National Parks, a tail bifurcation was observed in a female *Ophisops elegans* in the Phrygian Valley, Afyonkarahisar Province. The specimen was caught by hand. Morphometric, biotic, and environmental data were recorded as follows: The snout-vent length (SVL) was 52.67 mm, tail length (TL) was 48.56 mm, and the bifurcated portion of the tail measured 9.14 mm. These morphometric measurements were taken using a digital caliper (0.01 mm precision). The body mass was 2.72 g, measured with a digital scale (0.01 g precision). At the time of observation, the ambient temperature was 28.4 °C, measured using a hygro-thermometer, while the lizard's body temperature was measured at 33.5 °C by inserting a probed thermometer into the cloaca (0.1 °C precision). The angle of the tail bifurcation was approximately 33° (Fig. 1).



Figure 1. Lateral view of a bifurcated tail in a lizard, showing the split tail morphology (Photograph by Çağatay Bayram)

Ophisops elegans is a species that exhibits sexual color dimorphism; males display more patterns and appear more vibrant colored compared to females (Stuart-Fox & Ord, 2004). The specimen was diagnosed as a female based on external morphology and confirmed through the absence of palpable hemipenial pockets (Başoğlu & Baran, 1977; Bülbül & Sarıkurt, 2022). After photographic documentation, the lizard was released back into its natural habitat (Fig. 2).

Tamar et al. (2013) previously reported a case of bifurcation in a male individual snake-eyed lizard in Israel. The subspecies *O. e. blanfordi* found in Israel differs from *O. e. macrodactylus*, which inhabits our study area (Bozkurt et al., 2022). Bifurcation has been reported in various species within the family *Lacertidae*. These include *Podarcis muralis* (Pola & Koleska, 2017), *Eremias yarkandensis* (Liang et al., 2024), *Teira dugesii* (Nunes & Carretero, 2024), *Lacerta agilis*, and *Zootoca vivipara*

(Kolenda et al., 2017). Furthermore, bifurcation has also been documented in *Darevskia clarkorum* and *Darevskia rudis*, species distributed across the Anatolian region (Bülbül & Sarıkurt, 2022). The bifurcation in our study represents a moderate case for a *lacertidae* (Table 1).



Figure 2. Image of a bifurcated-tailed *Ophisops elegans* individual (Photograph by Muammer Kurnaz)

Table 1. Tail bifurcation lengths in *Lacertidae* species.

Study and species	SVL (Snout-vent Length) (mm)	TL (Tail Length) (mm)	Bifurcated Portion Length (mm)
<i>Ophisops elegans</i> (This study)	52.67	48.56	9,14
<i>Eremias yarkandensis</i> (Liang et al., 2024)	48.40	63.32	19,2
<i>Darevskia clarkorum</i> (Bülbül & Sarıkurt, 2022)	-	-	8,45
<i>Darevskia rudis</i> (Bülbül & Sarıkurt, 2022)	83.56	128.93	46,77

In conclusion, the observation of a bifurcated tail in *O. elegans* from Afyonkarahisar contributes to the growing body of evidence documenting partial tail autotomy and regeneration anomalies in lizards. This case adds to previous records within the family *Lacertidae* and underscores the morphological variability that can arise following autotomy, a process generally seen as an adaptive anti-predator defense. Given that tail bifurcation may have implications for locomotion, social behavior, or vulnerability to predation, such anomalies need further investigation. Continued field-based documentation of these occurrences, particularly across different subspecies and regions, can enhance our understanding of post-autotomy regeneration patterns and their ecological consequences.

Ethics committee approval: Ethics committee approval is not required for this study.

Conflict of interest: The authors declare that there is no conflict of interest.

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